

System of linear homogeneous eqⁿs :-

The given system of eqⁿs

$$a_{11}x + a_{12}y + a_{13}z = 0$$

$$a_{21}x + a_{22}y + a_{23}z = 0$$

$$a_{31}x + a_{32}y + a_{33}z = 0$$

i.e. $AX = 0$, where 0 is null column matrix is called ^① system of homogeneous eqⁿs

* If $|A| \neq 0$ i.e. A is non-singular, then from ① we get,

$$\text{①} \Rightarrow (A^{-1}A)X = A^{-1}0$$

$$IX = 0$$

$$\therefore X = 0 \text{ i.e. } X = 0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

then $x = y = z = 0$

∴ If $|A| \neq 0$, then above system has trivial solⁿ $x = y = z = 0$

* If $|A| = 0$ i.e. A is singular matrix then the system ① has non trivial solⁿ & homogeneous system is always consistent.

if $|A| = 0$ i.e. rank of $A < 3 (=n)$ then the system have $(n-r)$ linearly independent solⁿ.

if no. of eqⁿ < no. of unknown $\Rightarrow$ system have infinitely many solⁿ

if no. of eqⁿ = no. of unknown $\Rightarrow$ & $|A| = 0 \Rightarrow$ system have non trivial solⁿ

① Solve $x+2y+3z=0$, $3x+4y+4z=0$, $7x+10y+12z=0$

$\Rightarrow AX=0$ is the matrix form,

$$\text{Now } |A| = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 4 & 4 \\ 7 & 10 & 12 \end{vmatrix} = 1(48-40) - 2(36-28) + 3(30-28) \\ = 8 - 16 + 6 = -2 \neq 0$$

$\therefore |A| \neq 0$, Matrix A is non singular

$\therefore$ system have trivial soln

i.e. $x=y=z=0$

② Solve $4x+2y+z+3w=0$, $6x+3y+4z+7w=0$

$$2x+y+w=0$$

$\Rightarrow$ matrix form is $AX=0$

$$A = \begin{bmatrix} 4 & 2 & 1 & 3 \\ 6 & 3 & 4 & 7 \\ 2 & 1 & 0 & 1 \end{bmatrix} \begin{matrix} R_2 \rightarrow 4R_2 - 6R_1 \\ R_3 \rightarrow 2R_3 - R_1 \end{matrix} \sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 10 & 10 \\ 0 & 0 & -1 & -1 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{10}R_2 \quad R_3 \rightarrow -1R_3$$

$$A \sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} R_3 \rightarrow R_3 - R_2 \end{matrix} \sim \begin{bmatrix} 4 & 2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 2 \quad \& \quad \text{no. of unknown } (n) = 4$$

$$\therefore \text{no. of independent soln is } n - \rho = 4 - 2 = 2$$

$\therefore$ given system is equivalent to

$$\begin{cases} 4x + 2y + z + 3w = 0 \\ z + w = 0 \end{cases} \Rightarrow \boxed{z = -w}$$

$$\therefore 4x + 2y - w + 3w = 0$$

$$4x + 2y + 2w = 0$$

$$2x + y + w = 0 \Rightarrow \boxed{y = -2x - w}$$

which gives infinite no. of soln.

* Find k for which the system of eqns

$$(3k-8)x + 3y + 3z = 0, \quad 3x + (3k-8)y + 3z = 0$$

$$(3k-8)x + 3y + (3k-8)z = 0$$

$\Rightarrow$ for the given system of eqns to have a non trivial soln.

$$\therefore |A| = 0$$

$$\begin{vmatrix} 3k-8 & 3 & 3 \\ 3 & 3k-8 & 3 \\ 3 & 3 & 3k-8 \end{vmatrix} = 0$$

$$(3k-8)[(3k-8)^2 - 9] - 3[9 - 3(3k-8)] + 3[9 - 3(3k-8)] = 0$$

$$(3k-8)[9k^2 - 48k + 64 - 9] - 3[9k - 24 - 9] + 3[9 - 9k + 24] = 0$$

$$(3k-8)(9k^2 - 48k + 55) - 3(9k - 33) + 3(-9k + 33) = 0$$

$$27k^3 - 144k^2 + 165k - 72k^2 + 384 - 495 - 27k + 99 - 27k + 99 = 0$$

$$27k^3 - 72k^2 - 198k + 252 = 0$$

$$9k^3 - 24k^2 - 66k + 84 = 0$$

$$3k^3 - 8k^2 - 22k + 28 = 0$$

$$\begin{array}{r} 3 \cdot 8 - 8 \cdot 4 - 22 \cdot 2 \\ 27 \quad 9 \quad 3 \\ \hline 24 - 32 - 44 + 28 \\ 27 \quad 9 \quad 3 \\ \hline = 24 - 96 - 396 + 252 \\ 27 \end{array}$$

$$(3k-8)[(3k-8)^2 - 9] - 3[9k - 24 - 9] + 3[9 - 9k + 24] = 0$$

$$(3k-8)^3 - 27k + 72 - 27k + 99 - 27k + 99 = 0$$

$$(3k-8)^3 - 81k + 270 = 0$$

$$27k^3 - 3 \cdot 9k^2 \cdot 8 + 3 \cdot 3k \cdot 64 - 512 - 81k + 270 = 0$$

$$27k^3 - 216k^2 + 576k - 512 - 81k + 270 = 0$$

$$27k^3 - 216k^2 + 495k - 242 = 0$$

$$3k^3 - 24k^2 + 55k - 2 = 0$$

$$\Rightarrow k = \frac{2}{3}, \frac{11}{3}, \frac{11}{3}$$

* Find the value of λ for which the eqⁿs

$$(\lambda-1)x + (3\lambda+1)y + 2\lambda z = 0$$

$$(\lambda-1)x + (4\lambda-2)y + (\lambda+3)z = 0$$

$$2x + (3\lambda+1)y + 3(\lambda-1)z = 0$$

are consistent & find the ratios of $x:y:z$ when λ has the smallest of these values. what happened when λ has the greatest of these values.

$\Rightarrow$ The given eqⁿs will be consistent if

$$\begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ \lambda-1 & 4\lambda-2 & \lambda+3 \\ 2 & 3\lambda+1 & 3(\lambda-1) \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{vmatrix} \lambda-1 & 3\lambda+1 & 2\lambda \\ 0 & \lambda-3 & -\lambda+3 \\ 2 & 3\lambda+1 & 3(\lambda-1) \end{vmatrix} = 0$$

$$C_3 \rightarrow C_3 + C_2$$

$$\begin{vmatrix} \lambda-1 & 3\lambda+1 & 5\lambda+1 \\ 0 & \lambda-3 & 0 \\ 2 & 3\lambda+1 & 6\lambda-2 \end{vmatrix} = 0$$

$$(\lambda-1)[(\lambda-3) \cdot 2(3\lambda-1) - 0] - (3\lambda+1)[0-0] + (5\lambda+1)[0-2(\lambda-3)] = 0$$

$$(\lambda-1) \cdot 2(\lambda-3)(3\lambda-1) - 0 - (5\lambda+1) \cdot 2(\lambda-3) = 0$$

$$2(\lambda-3)[(\lambda-1)(3\lambda-1) - (5\lambda+1)] = 0$$

$$(\lambda-3)[3\lambda^2 - \lambda - 3\lambda + 1 - 5\lambda - 1] = 0$$

$$(\lambda-3)[3\lambda^2 - 9\lambda] = 0$$

$$(d-3) \cdot 3d(d-3) = 0$$

$$d(d-3)^2 = 0$$

$$\therefore d=0 \text{ or } d=3$$

Page No.

Date

When $d=0$ the eqn becomes

$$-x + y = 0 \Rightarrow x = y \quad \text{--- ①}$$

$$-x - 2y + 3z = 0 \quad \text{--- ②}$$

$$2x + y - 3z = 0 \quad \text{--- ③}$$

$$\frac{x}{-2/3} = \frac{-y}{-1/3} = \frac{z}{-1/2}$$

$$\frac{x}{6-3} = \frac{-y}{3-6} = \frac{z}{-1+4}$$

$$\frac{x}{3} = \frac{y}{3} = \frac{z}{3}$$

$$\Rightarrow x = y = z$$

$$\therefore x : y : z = 1 : 1 : 1$$

$$\text{②} \Rightarrow -x - 2x + 3z = 0$$

$$-3x + 3z = 0 \therefore x = z$$

$$\text{③} \Rightarrow 2z + y - 3z = 0$$

$$-z + y = 0$$

$$y = z$$

$$\therefore x = y = z$$

$$\therefore x : y : z = 1 : 1 : 1$$

When $d=3$, the eqn become identical

$$\text{i.e. } 2x + 10y + 6z = 0 \Rightarrow x + 5y + 3z = 0$$

$$2x + 10y + 6z = 0 \Rightarrow x + 5y + 3z = 0$$

$$2x + 10y + 6z = 0 \Rightarrow x + 5y + 3z = 0$$

$$\therefore x + 5y + 3z = 0$$

System have infinitely many soln.

Practice problems;

$$\text{① } x + y - 2z + 3w = 0, x - 2y + z - w = 0, 4x + y - 5z + 8w = 0, 5x - 7y + 2z - w = 0$$

$$\text{② } 3x + 4y - z - 6w = 0, 2x + 3y + 2z - 3w = 0, 2x + y - 14z - 9w = 0, x + 3y + 13z + 3w = 0$$

③ Determine the value of d for which the foll. sets of eqn may possess non trivial soln. $3x + y - dz = 0, 4x - 2y - 3z = 0, 2dx + 4y + dz = 0$. For each value of d find general soln.

* Characteristic eqⁿ, Eigen value & Eigen vector

Page No

Date

Characteristic Matrix $\Rightarrow$ If A is a square matrix of order n then the matrix $[A - \lambda I]$ is called as characteristic matrix

Characteristic eqⁿ $\Rightarrow$ The determinant of this matrix if equated to zero i.e. $|A - \lambda I| = 0$ is called as characteristic eqⁿ.

Eigen value $\Rightarrow$ The roots of characteristic eqⁿ are called eigen values or latent roots or characteristic roots.

~~Eigen vector~~

Eigen vectors $\Rightarrow$ If $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [x_1, x_2, \dots, x_n]^T$

is the non-zero column vector such that $AX = \lambda X$ or $[A - \lambda I]X = 0$ then X is called eigen vector corresponding to eigen value λ .

Properties of Eigen values $\Rightarrow$

- i) Sum of the eigen values of a matrix is equal to trace of a matrix (sum of diagonal elements)
- ii) If $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigen values of A then $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$ be the eigen values of A^m
- iii) If λ is an eigen value of A then $\frac{1}{\lambda}$ be the eigen value of A^{-1}

iv) Any sq. matrix A & its transpose A' have same eigen value.

v) eigen values of diagonal matrix or UTM or LTM are the diagonal elements of matrix

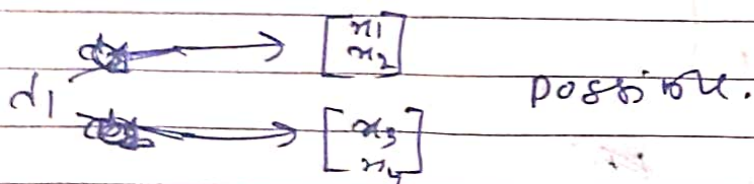
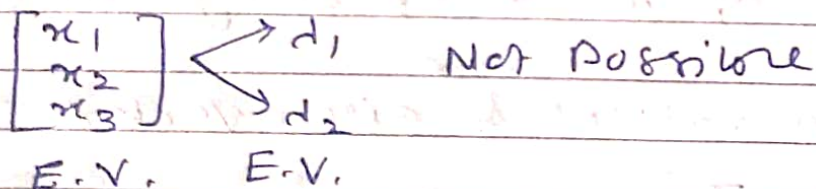
vi) If λ is eigen value of orthogonal matrix, then $\frac{1}{\lambda}$ is also eigen value.

vii) The E.V. of idempotent matrix are either 0 or 1. [Idempotent matrix $\Rightarrow A^2 = A$]

v) Product of eigen values of matrix A is equal with determinant of A
i.e. $|A| = d_1 \times d_2 \times d_3 \times \dots \times d_n$

Properties of eigen vectors:

i) corresponding to a given eigen vector there can not correspond two different eigen value but corresponding to a given eigen value there corresponds different eigen vectors.



2) corresponding to n distinct eigen values, we get n independent eigen vectors. But when two or more eigen values are equal, it may or may not be possible to get linearly independent eigen vectors corresponding to repeated roots.

3) Any sq. matrix A & its transpose A' have the same eigen

1) Find eigen values

Short-cut for characteristic eqn's

1) If A is a sq. matrix of order 2 then its characteristic eqn is

$$\lambda^2 - S_1\lambda + |A| = 0$$

2) If A is a sq. matrix of order 3 then its characteristic eqn is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

where $S_1 = \text{sum of diagonal elements}$

$S_2 = \text{sum of minor of diagonal elements}$

e.g. ① Find eigen values & eigen vectors of matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

Soln $\Rightarrow A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ ~~A'~~

Its characteristic eqn is

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0$$

$$S_1 = 1 + 5 + 1 = 7$$

$$S_2 = \begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = 4 - 8 + 4 = 0$$

$$|A| = 1(5-1) - 1(1-3) + 3(1-15) = 4 + 2 - 42 = -36$$

$$\therefore \lambda^3 - 7\lambda^2 + 0\lambda + 36 = 0$$

$$\lambda^3 - 7\lambda^2 + 36 = 0$$

$$\boxed{\lambda = -2, 3, 6}$$

Page No.

Date

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigen vector such that

$$AX = \lambda X$$

$$[A - \lambda I]X = 0$$

$$\begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(1-\lambda)x_1 + x_2 + 3x_3 = 0 \quad \text{--- (1)}$$

$$x_1 + (5-\lambda)x_2 + x_3 = 0 \quad \text{--- (2)}$$

$$3x_1 + x_2 + (1-\lambda)x_3 = 0 \quad \text{--- (3)}$$

For $\lambda = -2$,

$$3x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 7x_2 + x_3 = 0$$

$$3x_1 + x_2 + 3x_3 = 0$$

we have only two eqns with 3-unknowns

i.e. $3x_1 + x_2 + 3x_3 = 0$

& $x_1 + 7x_2 + x_3 = 0$

$$\frac{x_1}{\begin{vmatrix} 1 & 3 \\ 7 & 1 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}} = k$$

$$\frac{x_1}{-20} = \frac{-x_2}{0} = \frac{x_3}{20}$$

$$\frac{x_1}{1} = \frac{x_2}{0} = \frac{x_3}{-1} = k$$

$$\therefore X_1 = \begin{bmatrix} k \\ 0 \\ -k \end{bmatrix} \text{ for } \lambda = -2 \text{ i.e. } X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \text{ for } \lambda = -2$$

For $\lambda = 3$,

$$-2x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$3x_1 + x_2 - 2x_3 = 0$$

Page No.	
Date	

$$\frac{x_1}{2} = \frac{-x_2}{1} = \frac{x_3}{3} = k$$

$$\frac{x_1}{-5} = \frac{-x_2}{-5} = \frac{x_3}{-5} = k$$

$$\frac{x_1}{1} = \frac{x_2}{-1} = \frac{x_3}{1} = k$$

$$x_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \text{ for } \lambda = 3$$

For $\lambda = 6$,

$$-5x_1 + x_2 + 3x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$3x_1 + x_2 - 5x_3 = 0$$

$$\frac{x_1}{-1} = \frac{-x_2}{1} = \frac{x_3}{3} = k$$

$$\frac{x_1}{4} = \frac{-x_2}{-8} = \frac{x_3}{4} = k$$

$$\frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{1} = k$$

$$x_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \text{ for } \lambda = 6$$

There are 3 distinct eigen values

$\therefore$ The vectors x_1, x_2, x_3 are linearly independent

② Find the eigen values & eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

Page No.	
Date	

Solⁿ: Matrix A is an upper triangular matrix

$\therefore$ The eigen values for it are the diagonal elements of matrix,

$$\therefore \lambda = 3, 2, 5$$

Let $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ be an eigen vector s.t.
 $[A - \lambda I] x = 0$

$$\begin{bmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$(3-\lambda)x_1 + x_2 + 4x_3 = 0$$

$$(2-\lambda)x_2 + 6x_3 = 0$$

$$(5-\lambda)x_3 = 0 \Rightarrow \boxed{x_3 = 0}$$

For $\lambda = 3$,

$$0x_1 + x_2 + 4x_3 = 0$$

$$0x_1 - x_2 + 6x_3 = 0$$

$$0x_1 + 0x_2 + 2x_3 = 0$$

$$\frac{x_1}{-1/6} = \frac{-x_2}{0/6} = \frac{x_3}{0/-1} = k$$

$$\frac{x_1}{-2} = \frac{x_2}{0} = \frac{x_3}{0} = k \Rightarrow \frac{x_1}{1} = \frac{x_2}{0} = \frac{x_3}{0} = k$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ for } \lambda = 3$$

For $\lambda = 2 \Rightarrow x_1 + x_2 + 4x_3 = 0$

$0x_1 + 0x_2 + 6x_3 = 0$

$0x_1 + 0x_2 + 2x_3 = 0$

$\Rightarrow x_3 = 0$

We have two eqns

$x_1 + x_2 + 4x_3 = 0$

$0x_1 + 0x_2 + x_3 = 0$

$\frac{x_1}{1} = \frac{-x_2}{1} = \frac{x_3}{0} = k$

$x_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ for $\lambda = 2$

For $\lambda = 5 \Rightarrow -2x_1 + x_2 + 4x_3 = 0$

$0x_1 - 3x_2 + 6x_3 = 0 \Rightarrow 0x_1 - x_2 + 2x_3 = 0$

$0x_1 + 0x_2 + 0x_3 = 0$

~~eq~~ i.e. $-2x_1 + x_2 + 4x_3 = 0$

$0x_1 - x_2 + 2x_3 = 0$

$\frac{x_1}{6} = \frac{-x_2}{-4} = \frac{x_3}{2} = k$

$\frac{x_1}{3} = \frac{x_2}{2} = \frac{x_3}{1} = k$

$x_3 = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ for $\lambda = 5$

x_1, x_2, x_3 are independent vectors.

Practice Problems $\Rightarrow$

Page No.

Date

1) Find sum & product of the

eigen values of $\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$ also find the eigen values of its inverse.

2) Find eigen values & eigen vectors of the matrix

i) $\begin{bmatrix} 4 & 3 \\ 2 & 3 \end{bmatrix}$

ii) $\begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$

iii) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

iv) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ v) $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

vi) $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ vii) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Caley-Hamilton Theorem

Stat: $\Rightarrow$ Every square matrix satisfy its own characteristic eqⁿ i.e. if the characteristic eqⁿ for n^{th} order sq. matrix A is

$$|A - \lambda I| = 0$$

$$(-1)^n \lambda^n + k_1 \lambda^{n-1} + \dots + k_n = 0$$

then, $\boxed{(-1)^n A^n + k_1 A^{n-1} + \dots + k_n = 0}$

① verify C-H thm & find it's A inverse.
 Also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as
 a linear polynomial in A . if $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

$$\Rightarrow A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

It's characteristic eqn is

$$\lambda^2 - S_1\lambda + |A| = 0$$

$$S_1 = 1+3 = 4 \quad \& \quad |A| = 3-8 = -5$$

$$\lambda^2 - 4\lambda - 5 = 0 \quad \text{--- (1)}$$

By C-H thm, A must satisfy it's
 characteristic eqn $\therefore$ eqn (1) become

$$\therefore \text{(1)} \Rightarrow A^2 - 4A - 5I = 0$$

$$A^2 - 4A - 5I = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$+ \begin{bmatrix} -9 & -16 \\ -8 & -17 \end{bmatrix}$$

$$+ \begin{bmatrix} -9 & -16 \\ -8 & -17 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} + \begin{bmatrix} -9 & -16 \\ -8 & -17 \end{bmatrix}$$

$$= 0$$

$$A^2 - 4A - 5I = 0 \quad \text{Hence C-H thm}$$

$$\Delta. A^2 - 4A - 5I = 0 \quad - (2)$$

mul. by A^{-1}

$$A - 4I - 5A^{-1} = 0$$

$$5A^{-1} = A - 4I$$

$$5A^{-1} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$5A^{-1} = \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$$

$$= A^3 [A^2 - 4A - 7I] + 11 [A^2 - \frac{1}{11}A - \frac{10}{11}I]$$

$$= A^3 [\underbrace{A^2 - 4A - 5I}_{\downarrow} - 2I] + 11 [\underbrace{A^2 - 4A - 5I}_{\downarrow} + 4A + 5I - \frac{1}{11}A - \frac{10}{11}I]$$

$$= A^3 [0 - 2I] + 11 [0 + 4A + 5I - A - 10I]$$

$$= -2A^3 + 43A + 45I$$

$$= -2A [A^2 - \frac{43}{2}A] + 45I$$

$$= -2A [A^2 - 4A - 5I + 4A + 5I - \frac{43}{2}A] + 45I$$

$$= -2A [0 - 8A^2 - 10A + 43A^2] + 45I$$

$$= 35A^2 + 35I = 35[A^2 + I]$$

$$= 35[A^2 - 4A - 5I + 4A + 5I + I]$$

$$= 35[0 + 4A + 6I]$$

$$35A^2 - 10A + 45I = 0$$

$$35 \left[A^2 - \frac{10}{35}A + \frac{45}{35}I \right] = 0$$

$$35A^2 - 10A + 45I - \frac{10}{35}A + \frac{45}{35}I = 0$$

$$A^2 - 4A - 5I + 4A + 5I - \frac{2}{7}A + \frac{9}{7}I = 0$$

$$0 + \frac{26A}{7} + \frac{44I}{7} = 0$$

$$26A + 44I = 0$$

$$13A + 22I = 0$$

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I = 0$$

$$A^3 [A^2 - 4A - 7I] + 11A^2 - A - 10I = 0$$

$$A^3 [A^2 - 4A - 5I - 2I] + 11A^2 - A - 10I = 0$$

$$A^3 [0 - 2I] + 11A^2 - A - 10I = 0$$

$$-2A^3 + 11A^2 - A - 10I = 0$$

$$-2A \left[A^2 - \frac{11}{2}A + \frac{1}{2}I \right] - 10I = 0$$

$$-2A \left[A^2 - 4A - 5I + 4A + 5I - \frac{11}{2}A + \frac{1}{2}I \right] - 10I = 0$$

$$-2A \left[0 + 4A + 5I - \frac{11}{2}A + \frac{1}{2}I \right] - 10I = 0$$

$$-8A^2 - 10A + 11A^2 - A - 10I = 0$$

$$3A^2 - 11A - 10I = 0$$

$$3 \left[A^2 - \frac{11}{3}A - \frac{10}{3}I \right] = 0$$

$$A^2 - 4A - 5I + 4A + 5I - \frac{11}{3}A - \frac{10}{3}I = 0$$

$$0 + \frac{A}{3} + \frac{5}{3}I = 0 \Rightarrow \boxed{A + 5I = 0}$$

(OR)

we divided polynomial $x^5 - 4x^4 - 7x^3 + 11x^2 - x - 10$ by $x^2 - 4x - 5$

$$\begin{array}{r} x^3 - 2x + 3 \\ x^2 - 4x - 5 \overline{) x^5 - 4x^4 - 7x^3 + 11x^2 - x - 10} \\ \underline{-x^5 + 4x^4 - 5x^3} \\ -2x^3 + 11x^2 - x - 10 \\ \underline{-2x^3 + 8x^2 + 10x} \\ 3x^2 - 11x - 10 \\ \underline{-3x^2 + 12x + 15} \\ x + 5 \end{array}$$

$$\frac{a}{b} = q + \frac{r}{b}$$

$$\frac{x^5 - 4x^4 - 7x^3 + 11x^2 - x - 10}{x^2 - 4x - 5}$$

$$= (x^3 - 2x + 3) + \frac{(x + 5)}{(x^2 - 4x - 5)}$$

$$\therefore x^5 - 4x^4 - 7x^3 + 11x^2 - x - 10 = (x^3 - 2x + 3)(x^2 - 4x - 5) + x + 5$$

$$\begin{aligned} \therefore A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10 &= (A^3 - 2A + 3)(A^2 - 4A - 5) \\ &\quad + (A + 5) \\ &= 0 + (A + 5) \\ &= A + 5 \end{aligned}$$

Practice problems :->

Page No.	
Date	

① Find A^{-1} of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ by C-H thm.

② Find characteristic eqⁿ of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \text{ \& hence compute } A^{-1}$$

Also find the matrix represented by
 $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$

③ using C-H thm find A^8 if $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

④ verify C-H thm \& find A^{-1}

(i) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 7 & -1 & 3 \\ 6 & 1 & 4 \\ 2 & 4 & 8 \end{bmatrix}$

(iv) $\begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$

⑤ If $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ evaluate A^{-1}, A^{-2}, A^{-3}

Vector space $\Rightarrow$

Consider a non empty set V of vectors where each vector has the same number of components. If for any two vectors a & b in V , ~~then~~ such that

i) $\alpha a + \beta b$ ($\alpha, \beta \in \mathbb{R}$) $\in V$ (linear combination)

ii) $a + b = b + a$

iii) $a + 0 = a$

iv) $a + (-a) = 0$

& v) $(a + b) + c = a + (b + c)$

then V is vector space.

Linear dependence & Independence $\Rightarrow$

The set of vectors $x_1, x_2, \dots, x_n$ are said to be linearly dependent, if there exists scalars $k_1, k_2, \dots, k_n$ not all zero such that

$$k_1 x_1 + k_2 x_2 + \dots + k_n x_n = 0 \rightarrow \text{null vector}$$

If all $k_i = 0$, $i = 1, 2, \dots, n$ then vectors are said to be linearly independent.

Dimension $\Rightarrow$ The max. number of linearly independent vectors in V is c/a dimension of V . It is denoted by $\dim V$

Basis $\Rightarrow$ A linearly independent set x in V consisting of a max. possible no. of vectors in V is called a basis for V . [OR] any largest possible set of independent vectors in V forms basis for V .

$\therefore$ no. of vectors of a basis for $V = \dim V$

Span $\Rightarrow$ The set of all linear combiⁿs of given vectors $a_1, a_2, \dots, a_p$ with the same no. of components is called the span of these vectors.

obviously, a span is a vector space.

If the given vectors $a_1, a_2, \dots, a_p$ are linearly independent then they form a basis for the vector space.

Subspace $\Rightarrow$ Let W be any nonempty subset of V & W forms a vector space w.r.t. the two algebraic operatⁿ (addiⁿ & scalar mul.) defined for the vectors of V

i.e. that for any vectors $a, b \in W$ s.t.

i) $\alpha a + \beta b \in W$ $\alpha, \beta = \text{scalars}$

ii) $a + b = b + a$

iii) $a + 0 = a$

iv) $a + (-a) = 0$

v) $(a + b) + c = a + (b + c)$

then W is called subspace of V .

① Are foll. sets of vectors linearly independent
 $[3 \ 4 \ 0 \ 1], [2 \ -1 \ 3 \ 4], [1 \ 6 \ -8 \ -2]$

Page No.	8
Date	

Solⁿ $\Rightarrow$ Let the linear combⁿ of above vectors is

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$$

$$c_1 [3 \ 4 \ 0 \ 1] + c_2 [2 \ -1 \ 3 \ 4] + c_3 [1 \ 6 \ -8 \ -2] = 0$$

$$\therefore 3c_1 + 2c_2 + c_3 = 0$$

$$4c_1 - c_2 + 6c_3 = 0$$

$$0c_1 + 3c_2 - 8c_3 = 0$$

$$c_1 + 4c_2 - 2c_3 = 0$$

$\therefore$ The matrix form is $AX = 0$

$\therefore$ The Augmented matrix is

$$[A:0] = \begin{bmatrix} 3 & 2 & 1 & : & 0 \\ 4 & -1 & 6 & : & 0 \\ 0 & 3 & -8 & : & 0 \\ 1 & 4 & -2 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow 3R_2 - 4R_1, \quad R_4 \rightarrow 3R_4 - R_1$$

$$[A:0] \sim \begin{bmatrix} 3 & 2 & 1 & : & 0 \\ 0 & -11 & 14 & : & 0 \\ 0 & 3 & -8 & : & 0 \\ 0 & 10 & -7 & : & 0 \end{bmatrix}$$

$$R_2 \rightarrow 11R_2 + 3R_3, \quad R_4 \rightarrow 11R_4 + 10R_2$$

$$[A:0] \sim \begin{bmatrix} 3 & 2 & 1 & : & 0 \\ 0 & -11 & 14 & : & 0 \\ 0 & 0 & -46 & : & 0 \\ 0 & 0 & 63 & : & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 2 & 1 & : & 0 \\ 0 & -11 & 14 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$R_3 \rightarrow \frac{1}{-46} R_3$
 $R_4 \rightarrow \frac{1}{63} R_4$

$$R_4 \rightarrow R_4 - R_3$$

$$[A:0] \sim \begin{bmatrix} 3 & 2 & 1 & : & 0 \\ 0 & -11 & 14 & : & 0 \\ 0 & 0 & 1 & : & 0 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow \frac{1}{11}R_3 - R_2$$

Page No.			
Date			

$$\therefore \rho(A;0) = \rho(A) = 3 = n$$

$$\therefore 3c_1 + 2c_2 + c_3 = 0 \quad \text{--- (1)}$$

$$-11c_2 + 14c_3 = 0 \quad \text{--- (2)}$$

$$\boxed{c_3 = 0}$$

$$\therefore (2) \Rightarrow \boxed{c_2 = 0}$$

$$\therefore \boxed{c_1 = 0}$$

$\therefore$ system has

$\therefore$ Given vectors are linearly independent.

$\therefore$ they form a basis

$$(2) [1, 0, 1], [1, 1, 0], [0, 1, 0]$$

Their linear combin is

$$c_1 a_1 + c_2 a_2 + c_3 a_3 = 0$$

$$c_1 [1, 0, 1] + c_2 [1, 1, 0] + c_3 [0, 1, 0] = 0$$

$$c_1 + c_2 + 0c_3 = 0$$

$$0c_1 + c_2 + c_3 = 0$$

$$c_1 + 0c_2 + 0c_3 = 0 \Rightarrow$$

$$\begin{cases} c_1 = 0 \\ c_2 = 0 \\ c_3 = 0 \end{cases}$$

$\Rightarrow$ vectors are linearly independent

OR

$\therefore$ They form a basis

The matrix form is $AX = 0$
 $\therefore$ The Augmented matrix is

Page No.			
Date			

$$[A:0] \sim \begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 1 & 0 & 0 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & -1 & 0 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 0 & : & 0 \\ 0 & 1 & 1 & : & 0 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$\rho(A) = \rho(A:0) = 3 = n$$

$\therefore$ System has unique solⁿ.

$$\begin{array}{l|l} c_1 + c_2 = 0 & \text{if } c_3 = 0 \\ c_2 + c_3 = 0 & \therefore c_2 = 0 \text{ \& } c_1 = 0 \\ c_3 = 0 & \end{array}$$

$\therefore$ all vectors are linearly independent.

$$(3) [4 \ -1 \ 3], [0 \ 8 \ 1], [1 \ 3 \ -5], [2 \ 6 \ 1]$$

$$c_1 a_1 + c_2 a_2 + c_3 a_3 + c_4 a_4 = 0$$

$$4c_1 + 0c_2 + c_3 + 2c_4 = 0$$

$$-c_1 + 8c_2 + 3c_3 + 6c_4 = 0$$

$$3c_1 + c_2 - 5c_3 + c_4 = 0$$

$\rightarrow (A)$

Here total
vectors = $p = 4$
no. of components
in each vector
 $n = 3$
 $\therefore n < p$
vectors are l.d.

$$[A:0] = \begin{bmatrix} 4 & 0 & 1 & 2 & : & 0 \\ -1 & 8 & 3 & 6 & : & 0 \\ 3 & 1 & -5 & 1 & : & 0 \end{bmatrix}$$

Page No.			
Date			

$$R_2 \rightarrow 4R_2 + R_1 \quad R_3 \rightarrow 4R_3 - 3R_1$$

$$[A:0] = \begin{bmatrix} 4 & 0 & 1 & 2 & : & 0 \\ 0 & 32 & 13 & 26 & : & 0 \\ 0 & 4 & -23 & -2 & : & 0 \end{bmatrix}$$

$$R_3 \rightarrow 8R_3 - R_2$$

$$= \begin{bmatrix} 4 & 0 & 1 & 2 & : & 0 \\ 0 & 32 & 13 & 26 & : & 0 \\ 0 & 0 & -197 & -42 & : & 0 \end{bmatrix}$$

$$\rho(A:0) = \rho(A) = 3 < n (=4)$$

$\therefore$ system have non trivial soln & they are infinite.

$\therefore$ vectors are l.d.

$$\therefore 4c_1 + c_3 + 2c_4 = 0 \quad \text{--- (1)}$$

$$32c_2 + 13c_3 + 26c_4 = 0 \quad \text{--- (2)}$$

$$-197c_3 - 42c_4 = 0 \quad \text{--- (3)}$$

$$(3) \Rightarrow 197c_3 + 42c_4 = 0$$

$$\text{let } c_3 = 1 \quad \left| \begin{array}{l} (2) \Rightarrow 32c_2 + 13 \times 1 + 26 \times \frac{197}{42} = 0 \\ 32c_2 + 13 + \frac{2561}{21} = 0 \end{array} \right. \quad \begin{array}{l} 13 \\ 42 \\ 21 \end{array}$$

$$32c_2 = -\frac{2834}{21}$$

$$c_2 =$$

$$\text{let } c_3 = K$$

$$c_4 = -\frac{197K}{42}$$

$$① \Rightarrow 4c_1 + c_3 + 2c_4 = 0$$

Page No.

Date

$$4c_1 + k - \cancel{197} \times \frac{197}{\cancel{42} \times 21} k = 0$$

$$4c_1 + k - \frac{197}{21} k = 0$$

$$4c_1 - \cancel{88} \frac{176}{21} k = 0$$

$$c_1 = \frac{\cancel{44} \frac{176}{21} k}{\cancel{4}} = \frac{44}{21} k$$

$$\boxed{c_1 = \frac{44}{21} k}$$

$$② \Rightarrow 32c_2 + 13c_3 + 26c_4 = 0$$

$$\frac{32}{13} c_2 + c_3 + 2c_4 = 0$$

$$\frac{32}{13} c_2 + k - \cancel{197} \times \frac{197}{\cancel{42} \times 21} k = 0$$

$$\frac{32}{13} c_2 + k - \frac{197}{21} k = 0$$

$$\frac{32}{13} c_2 = \frac{176}{21} k$$

$$c_2 = \frac{\cancel{1144} \frac{176}{21} k}{\cancel{32} \times 82} = \frac{143}{42} k$$

$$\boxed{c_2 = \frac{143}{42} k}$$

$$\text{let } k = 42$$

$$\therefore c_3 = 42, c_2 = 143, c_1 = 88, c_4 = -197$$

$$⑤ \Rightarrow 88a_1 + 143a_2 + 42a_3 - 197a_4 = 0$$

Practice Problems : $\Rightarrow$

Page No.

Date / /

Are the foll. vectors linearly dependent?

If so find the relⁿ betⁿ them

① $(2, 1, 1), (2, 0, -1), (4, 2, 1)$

② $(1, -1, 0), (0, 1, -1), (0, 0, 1)$

③ $(1, 1, -1, 1), (1, -1, 2, -1), (3, 1, 0, 1)$

④ $(1, 2, 4), (2, -1, 3), (0, 1, 2), (-3, 7, 2)$

⑤ $(6, 0, -1, 3), (2, 2, 5, 0), (-4, -4, -4, -4)$

⑥ $(1, 2, 3, 4), (2, 3, 4, 5), (3, 4, 5, 6), (4, 5, 6, 7)$

⑦ $(9, 8, 7, 6, 5), (9, 7, 5, 3, 1)$

⑧ $(2, 0, 0, 7), (2, 0, 0, 8), (2, 0, 0, 9), (2, 0, 1, 0)$

⑨ $(1, 2, 1), (2, 1, 4), (4, 5, 6), (1, 8, -3)$

⑩ $(2, 1), (3, 2), (1, 1), (2, 3)$

⑪ $(2, -1, 4), (0, 1, 2), (6, 1, 14), (4, 0, 12)$

⑫ $(3, 2, 4), (1, 0, 2), (1, -1, 1)$

⑬ Express the vector $(3, 7, -4) \in \mathbb{R}^3$ as a linear combinⁿ of $x_1 = (1, 2, 3), x_2 = (2, 3, 2), x_3 = (3, 5, 6)$

⑭ Express the vector $(2, 3, -1, -1) \in \mathbb{R}^4$ as a linear combinⁿ of $(3, 1, 3, 2), (1, -1, -2, -4), (6, 3, 0, -7)$

Linear Transform

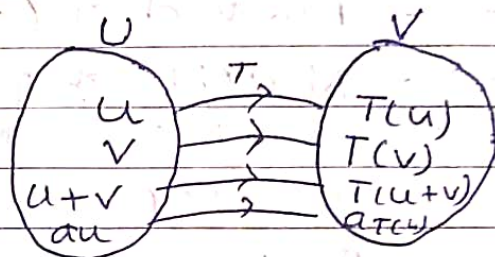
Let U & V be two vector spaces over the same field F . Then a map (fun.) $T: U \rightarrow V$ is called linear transform if

- i) $T(u+v) = T(u) + T(v)$, $\forall u, v \in U$
 ii) $T(au) = aT(u)$, $\forall a \in F, u \in U$

OR

$$T(au + bv) = aT(u) + bT(v)$$

Properties:



i) $T(\vec{0}_U) = \vec{0}_V$

ii) $T(-u) = -T(u)$

iii) $T(u-v) = T(u) - T(v)$

Null space or kernel of T :

Let $T: U \rightarrow V$ be a linear transform. The kernel of T is subset of U consisting of all vectors u whose image under T is $\vec{0}_V$.

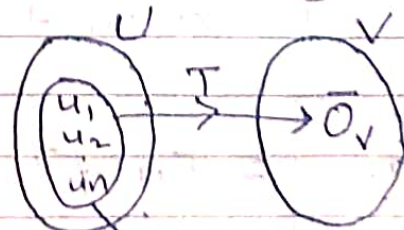
i.e. $\ker T = N(T) = \{u \in U / T(u) = \vec{0}_V\}$

Note:

$N(T)$ is subspace of U

i.e. $N(T)$ is nonempty &

$u, v \in N(T) \Rightarrow au + bv \in N(T)$

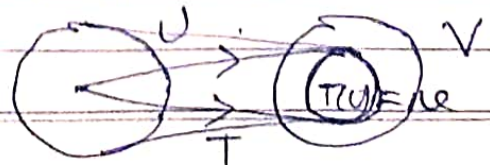


$N(T)$ or $K(T)$

Range of T : Let $T: U \rightarrow V$ be linear transform.

The range/image of T is the set of all vectors in V that are image under T of vectors in U .

Range of $T = R(T) = \{T(u) / u \in U\}$



Rank & Nullity of a linear transformation

$$\left. \begin{array}{l} R(T) = \{ T(u) \mid u \in U \} \\ \downarrow \\ \text{subspace of } V \\ N(T) = \{ u \in U \mid T(u) = 0_V \} \\ \downarrow \\ \text{subspace of } U \end{array} \right\} \rightarrow \text{subspace (finite)} \quad \therefore \text{they form a basis.}$$

→ no. of ele. in basis of $R(T)$

$$\dim R(T) = \text{Rank of } T (S(T))$$

$$\dim N(T) = \text{Nullity of } T (N(T))$$

→ no. of ele. in basis of $N(T)$

Thm: If $T: U(F) \rightarrow V(F)$ is a linear transformation then $\text{Rank } T + \text{Nullity } T = \dim U$

$$\text{Rank } T + \text{Nullity } T = \dim U$$

i.e. $S(T) + N(T) = \dim U$

Coordinate Vector: Let $V(F)$ is finitely generated vector space.

$\therefore$ basis of $V(F)$ is $B = \{u_1, u_2, \dots, u_m\}$

& let $u \in V$ then we can write u as a linear combiⁿ of $\{u_1, u_2, \dots, u_m\} = B$

$$\text{i.e. } u = a_1 u_1 + a_2 u_2 + a_3 u_3 + \dots + a_m u_m$$

$$\therefore \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} \equiv [u, B] \text{ is coordinate vector.}$$

→ u as a l.c. of B

matrix of linear transformation relative to
order basis:-

$$T: U \rightarrow V$$

let $B = \{u_1, u_2, \dots, u_n\}$ basis of $U \subseteq U$
& $B' = \{v_1, v_2, \dots, v_m\}$ —, —, — $V \subseteq V$

also $T(u_1), T(u_2), \dots, T(u_n) \in V$
 $\therefore$ we can write $T(u_1)$ as a L.C. of B'

$$T(u_1) = a_{11}v_1 + a_{21}v_2 + \dots + a_{m1}v_m$$

$$T(u_2) = a_{12}v_1 + a_{22}v_2 + \dots + a_{m2}v_m$$

$$T(u_n) = a_{1n}v_1 + a_{2n}v_2 + \dots + a_{mn}v_m$$

$$\begin{bmatrix} a_{11} & a_{21} & \dots & a_{m1} \\ a_{12} & a_{22} & \dots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{mn} \end{bmatrix}_{n \times m}$$

Take Transpose of above matrix

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n} = \text{matrix of L.T. relative to order basis.} = [T; B, B'] \leftarrow$$

① Find linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ whose matrix is
 $A = \begin{bmatrix} 1 & -1 \\ -2 & 3 \\ 0 & 1 \end{bmatrix}$, $B = \{(1, 1), (0, 2)\}$
 $B' = \{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$

Solⁿ: $\Rightarrow$ Given map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$
 $(u) \rightarrow (v)$

$\therefore$ elements of $B \in U \subseteq \mathbb{R}^2$ & ele. of $B' \in V = \mathbb{R}^3$

$$\therefore T(u_i) = a_{1i}v_1 + a_{2i}v_2 + a_{3i}v_3$$

$$T(1, 1) = 1(0, 1, 1) + (-2)(1, 0, 1) + 0(1, 1, 0)$$

$$T(1, 1) = (0, 1, 1) - (2, 0, 2) + 0$$

$$\boxed{T(1, 1) = (-2, 1, -1)}$$

$$\text{ii) } T(0, 2) = -1(0, 1, 1) + 3(1, 0, 1) + 1(1, 1, 0)$$

$$\downarrow$$

$$\boxed{T(0, 2) = (4, 0, 2)}$$

To find linear map i.e. to find $T(x, y)$
 i.e. $T(x, y) = \left(\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right)$
 Return n by

Let $(x, y) \in \mathbb{R}^2$ $\therefore$ we can write (x, y) as
 L.C. of B .

$$(x, y) = a(1, 1) + b(0, 2) \quad \text{--- (1)}$$

$$(x, y) = (a, a + 2b) \Rightarrow \boxed{x = a} \text{ \& } \boxed{b = \frac{y-a}{2} = \frac{y-x}{2}}$$

$$\text{Now, } T(x, y) = T[a(1, 1) + b(0, 2)] \quad \text{--- (2)}$$

$$= aT(1, 1) + bT(0, 2)$$

$$= a(-2, 1, -1) + b(4, 0, 2)$$

$$T(x, y) = x(-2, 1, -1) + \left(\frac{y-x}{2}\right)(4, 0, 2) \\ = (-2x, x, -x) + (2y-2x, 0, y-x)$$

$$T(x, y) = (2y-4x, x, y-2x)$$

② Find matrix representing $T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$ such that $T(x, y, z) = (x+y+z, 2x+z, 2y-z, 6y)$ relative to std. basis of $\mathbb{R}^3$ & $\mathbb{R}^4$.

$$\Rightarrow \text{std. basis of } \mathbb{R}^3 = B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \\ \text{std. basis of } \mathbb{R}^4 = B' = \{(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)\} \\ \downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ e_1 \quad e_2 \quad e_3 \quad e_4$$

$$\begin{aligned} T(1, 0, 0) &= (1, 2, 0, 0) \in \mathbb{R}^4 \\ T(0, 1, 0) &= (1, 0, 2, 6) \\ T(0, 0, 1) &= (1, 1, -1, 0) \end{aligned} \left. \vphantom{\begin{aligned} T(1, 0, 0) \\ T(0, 1, 0) \\ T(0, 0, 1) \end{aligned}} \right\} \in \mathbb{R}^4$$

$$T(1, 0, 0) = (1, 2, 0, 0) = 1e_1 + 2e_2 + 0e_3 + 0e_4$$

$$T(0, 1, 0) = (1, 0, 2, 6) = 1e_1 + 0e_2 + 2e_3 + 6e_4$$

$$T(0, 0, 1) = (1, 1, -1, 0) = 1e_1 + 1e_2 - 1e_3 + 0e_4$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 1 & 0 & 2 & 6 \\ 1 & 1 & -1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & -1 \\ 0 & 2 & 0 \\ 0 & 6 & 0 \end{bmatrix} = [T: B, B']$$